Midterm Project: Classifying Text Using CNN and RNN

Problem Definition

Since ChatGPT's debut in 2022, LLM usage has skyrocketed. Fast-forward to 2026, the share of LLMs is split between many chatbots, such as ChatGPT, Claude, Gemini, and Llama. The widespread usage of LLMs has produced the need to determine whether a piece of text was produced by a human or a chatbot. Furthermore, knowing which LLM produced a text is a growing topic.

Machine learning is used in many classification problems. This includes predicting what an image is, producing recommendations given user preferences, and determining health risks given a person's lifestyle. Two deep-learning architectures, convolutional neural networks (CNN) and recurrent neural networks (RNN), could be useful for classifying LLMs given their outputs.

Convolutional neural networks convolve a filter around the input to produce activation maps, allowing a machine learning model to split an input into subparts with shared parameters. They are quick and can perceive context in both directions, but struggle to perceive context in lengthy inputs.

Recurrent neural networks apply a recurrence formula to a sequence of vectors many times, which allows a model to recognize longer context. However, RNNs can only perceive context on one side and are very slow.

The goal is to determine the best approach, CNN or RNN, for determining which LLM produced a piece of text. It is also to analyze whether the user's prompt can also aid in the classification of LLM outputs.

Dataset Curation

The dataset used was the LMSYS Chatbot Arena Conversations Dataset. The dataset was collected from the Chabot Arena from April to June 2023, where users created a prompt to be presented to two randomly selected LLMs and selected the better LLM output. This crowdsourced project was meant to help understand human preferences in AI outputs and to rank models. This dataset can be useful for tuning LLMs for human preference, allowing for better outputs.

The LLMs included in this dataset were:

- RWKV 4 Raven 14B
- Alpaca 13B
- ChatGLM 6B
- Claude Instant V1
- Claude V1
- Dolly V2 12B
- FastChat T5 3B
- GPT 4
- GPT4ALL 13B Snoozy
- Guanaco 33B
- Koala 13B
- Llama 13B
- MPT 7B Chat
- Oasst Pythia 12B
- Palm 2
- Stablelm Tuned Alpha 7B
- Viscuna 13B
- Vicuna 7B
- Wizardlm 13B

User information was anonymized in this dataset, so there should not have been any identifying information used in this project. The columns used in this dataset include:

- “model_a” and “model_b”, which identify the LLMs used
- “conversation_a” and “conversation_b”, which are the user prompts and LLM outputs for the respective LLM in a conversation.

The data was transformed into the format

- “source”, which describes the LLM that produced the output
- “prompt_name”, which describes the user’s prompt
- “text”, which describes the LLM’s output

Only the first prompt and first LLM output of the conversations were used, as the following messages may have used context from previous messages which would not be able to be captured in this format. The CNN and RNN were trained on multiple versions of the dataset which limited classes of LLMs. This included:

- All LLMs as separate classes
- Only 3 LLMs from this dataset, being GPT 4, Claude V1, and Palm 2
- Grouped into their respective families which would be their class
- Only 3 families of LLMs, being GPT, Claude, Llama

Results

RQ1 — Can we identify which LLM generated a response? Use LLM_output as input, conduct classification to detect which LLM generated it. Report the performance of your CNN&RNN performance

When the LLMs were limited to just GPT 4, Claude V1, and Palm 2, the CNN had an accuracy of 74% and the RNN had an accuracy of 67%. The CNN had the strongest f1 score of 0.80 with Palm 2, while the RNN had the strongest f1 score of 0.78 with Claude V1.

When provided with all 19 LLMs as separate classes, both models performed poorly. The CNN was only 19% accurate and the RNN was 18% accurate. This is likely caused by LLMs having similar writing styles or even being part of the same family, making them hard to distinguish.

Therefore, we could identify which LLM generated a response using a CNN or RNN, but only if the possible sources are limited down to a few distinguishable LLMs first.

RQ2 — Does the user's prompt help identify the LLM? Compare classifiers using: Input only; Output only; Input + Output. Questions you may consider: i) Can the LLM be predicted from the user's prompt even before seeing its response? ii) Does including the input improve classification? iii) If input-only classification works surprisingly well, why might this happen?

i. The LLMs could not identify the LLM with just the input. When the LLMs were limited to just GPT 4, Claude V1, and Palm 2, the CNN had an accuracy of 39% and the RNN had an accuracy of 38%. Both the CNN and RNN had the strongest f1 scores with GPT 4. Ultimately, you cannot accurately predict the LLM with just the input.

ii. When provided both the input and output, the CNN had slightly better test accuracy, with 76% and f1 scores of 0.76 across all 3 LLMs. However, the RNN had a worse accuracy with both input and output, having only 63% accuracy, with the highest f1 score of 0.70 on Claude V1. Overall, the user's prompt could help identify the LLM, but only in the CNN and not by much.

iii. If the input-only classification works surprisingly well, then it might indicate there might be a disproportionate amount of data from only one LLM, and the model is just guessing the most common LLM, resulting in high test scores. This might also occur if the dataset contains language models with completely different use cases, such as one for coding, one for text summarization, and one for solving math problems. As a result, the user inputs would be more distinct and could be used to identify the LLM used.

In-Depth Analysis and Experiments

The CNN performed better than the RNN with lower testing loss whenever the models could properly infer the correct LLM. This could be caused by CNN's bi-directional context perception and shorter LLM outputs which reduces the advantage of RNN. Training in each epoch was noisier in the RNN compared to the CNN, and it was much slower since training in each epoch was slow and more epochs were needed before early stopping could take place.

Adding user prompts only slightly improved the CNNs classification and did not seem to affect the RNNs performance, apart from making training slower as it had to train on longer inputs. In this dataset, the same input was used on at least two different LLMs, and the LLM was chosen randomly, which could potentially lessen the usefulness of the user's input.

If the CNN and RNN model is trained on all 19 LLMs, where each LLM represents its own class, the performance of both models is poor. This could be caused by multiple LLMs having similar writing styles but are separate classes. The most distinguishable LLM was Claude V1, followed by Palm, as both had the highest f1 scores for output-only and input+output in both the CNN and the RNN. This may convey that these two models have distinguishing writing characteristics that make them identifiable compared to the other LLMs.

Grouping the LLMs by families slightly significantly increases performance. Both the RNN and the CNN had strong performance with identifying outputs and inputs+outputs which belonged to the Claude family or the Llama family. The CNN had higher f1 scores for Palm 2 in output-only training and evaluation compared to the rest of the families, but higher f1 scores for the GPT family for input+output. The RNN had high f1 scores for the GPT family in output only. Both the CNN and RNN struggled to classify the rest of the families.

Limiting families to the Claude, GPT, and Llama families further increased performance. Overall, both struggled with identifying the GPT family, but the CNN had higher f1 scores for GPT compared to the RNN.

Model Accuracies

	CNN Output	CNN Input	CNN Input+Output	RNN Output	RNN Input	RNN Input+Output
All LLMs	0.19	0.09	0.18	0.18	0.09	0.15
LLMs grouped by families	0.42	0.36	0.41	0.43	0.36	0.41
GPT 4, Claude V1, and Palm 2	0.74	0.39	0.76	0.67	0.38	0.63
GPT family, Claude family, and Llama Family	0.68	0.61	0.67	0.68	0.61	0.70

Lessons

Overall, this project highlights the key strengths and weaknesses of CNNs and RNNs. CNNs are faster and can see context in both directions, while RNNs trade speed and bi-directional context for longer context. In this case, where LLM outputs are no more than a few hundred words, so a CNN would be a better choice.

It also highlights how LLMs can have distinguishable writing styles, which can be used to identify if it wrote a specific piece of text and could be detected via machine learning. It also shows the drawbacks of classification models when these distinguishing features are hard to find between multiple LLMs, resulting in poor classification results. This is shown when similar LLMs are their own class.